Hello, I am Debayan Mitra, a fresher in electrical engineering (dual degree) and I am really excited to begin my journey in semiconductor fabrication and studying it in detail.
My introduction to robotics after joining IIT Bombay began through XLR8, where my team (I built the servo and ultrasonic parts) **won the MRT problem statement,** an experience that taught me the fundamental mechanics of how machines move and operate. However, I have always felt that **true hardware engineering extends far beyond mechanics which requires mastering the silicon and semiconductor devices that control them**. For me, HackerFab is an **opportunity to do active hands-on work to design tube furnaces, construct custom processing tools, and work directly within a unique, machinery driven space on campus**. Ever since I began exploring the semiconductor industry during my school years, I have been **deeply fascinated by how sophisticated, precise, and elegant the entire manufacturing pipeline is**. Watching **Veritasium's deep dive into EUV lithography** further opened my eyes to the **extraordinary engineering scale and technical rigor involved in modern microelectronics**. I am drawn to this domain because it **sits at the intersection of almost every major engineering discipline**, from materials science and mechanical design to electrical engineering offering me an **ideal technical foundation to build, experiment, and solve complex problems across multiple domains**.Since choosing engineering, my overarching goal has been to **work at the absolute frontier of technology, particularly in semiconductor fabrication and device engineering**, and HackerFab represents the ideal starting point for this journey. During my lab visit (thanks to Rushat :)) I learnt a lot about how a student run tech lab works, this further motivated me to join the team as it presents an **opportunity to use cutting edge machinery** and take apart and study many more, what fascinated me the most was the **repurposing of various machines** and how innovative and scrappily things were built in the lab. In the short term, I **aim to bridge theoretical device physics with practical cleanroom execution** by helping build a **closed end tube furnace** for in-house thermal oxidation and spin on dopant drive in. Achieving precise environment control within the furnace will allow us to reliably dope silicon wafers, paving the way for fabricating functional Enhancement-Mode MOSFETs and complex integrated circuits, including the CMOS ring inverter I proposed in problem statement A. Long term, I aspire to drive the team's tool building capabilities forward by constructing an **RF AC-sputter coater** for thin-film deposition, engineering its **automated control systems**, and **applying robotics principles to lithography** substrate alignment, ensuring our team continues to push the envelope of accessible microfabrication**.** To conclude, my goal has always been to work at the absolute frontier of technology, and true progress starts at the silicon level. HackerFab offers the ideal environment to bridge theory and physical engineering, where I can get my hands dirty constructing furnaces, automating lithography robotics, and repurposing hardware. I am eager to bring my curiosity, scrappiness, and technical drive to the team to continue to progress student run nanofabs across the globe.